Questions

Q1.

This question is about radioactivity.

Figure 14 shows a Geiger-Muller (G-M) tube attached to a counter.
The G-M tube is used to measure the activity of a source of beta (β) radiation.
There is an aluminium sheet between the beta source and the G-M tube.
The counter is switched on and after 1 minute shows a count of 268.

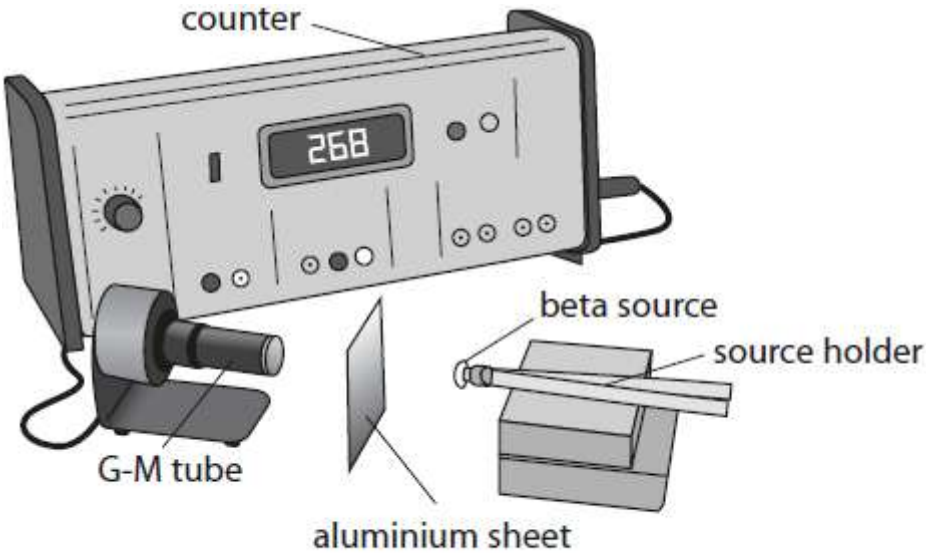

Figure 14

(i) The aluminium sheet is taken away.
The counter is reset to zero and then switched on again.
A new count is taken for 1 minute.

Explain why the new count is greater than 268.

(2)

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(ii) The beta source is then also taken away.
The counter is reset to zero and switched on again.
A new count is taken for 1 minute.

Give a reason why there would now be a reading on the counter.

(1)

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(iii) State the SI unit for the activity of a radioactive source.

(1)

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(Total for question = 4 marks)

Q2.

Uranium-238 can only undergo nuclear fission by absorbing fast neutrons.
The fission emits neutrons which very quickly lose their energy.

Suggest why the fission of uranium-238 does not produce a chain reaction.

(2)

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Q3.

Figure 8 shows two types of force that act in the Sun.

One type of force acts inwards, towards the centre of the Sun.

The other type of force acts outwards, away from the centre of the Sun.

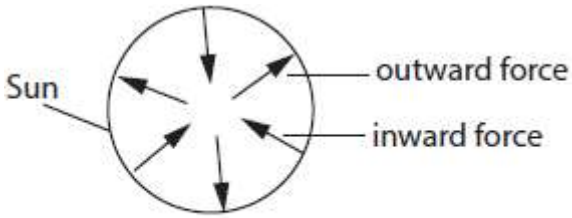

Figure 8

In 1989, two scientists claimed that they could produce a nuclear fusion reaction in their laboratory.

They said their reaction could take place at room temperature in a test tube.

Explain why other scientists did not believe them.

(2)

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(Total for question = 2 marks)

Q4.

*Figure 14 shows a household smoke alarm that uses radioactivity to detect smoke.

Courtesy NASA/JPL-Caltech

Figure 14

The radioactive source in the smoke detector is americium–241.

The table in Figure 15 shows some information about americium–241 and two other radioactive sources.

radioactive source	type of radiation	half-life
americium–241	alpha	433 years
actinium-225	alpha	10 days
cobalt-60	gamma	5.27 years

Figure 15

Explain why americium-241 is the best of these three sources to use in this smoke detector.

Use information from Figure 15 and your own knowledge about radiation.

Your answer should refer to

- properties of alpha and gamma radiation
- half-life. **(6)**

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(Total for question = 6 marks)

Q5.

In 1908 a scientist called Rutherford was investigating ideas about atoms.

His students fired a beam of alpha particles at a thin piece of gold foil.

Figure 10 shows the arrangement of the experiment.

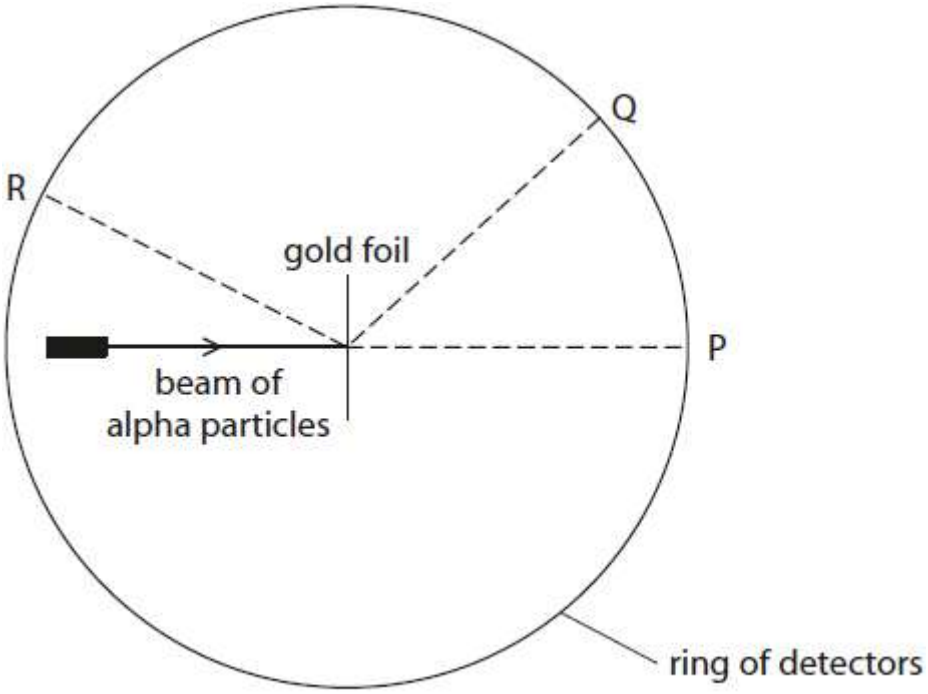

Figure 10

Some alpha particles were found at all parts of the ring of detectors.

The table in Figure 11 shows how many alpha particles were detected at P, at Q and at R, in one experiment.

position	number of alpha particles detected
P	72340
Q	25
R	2

Figure 11

Explain what the information in Figure 10 and Figure 11 shows about the structure of an atom.

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(Total for question = 6 marks)

Q6.

Figure 21 shows what happens during a fission chain reaction.

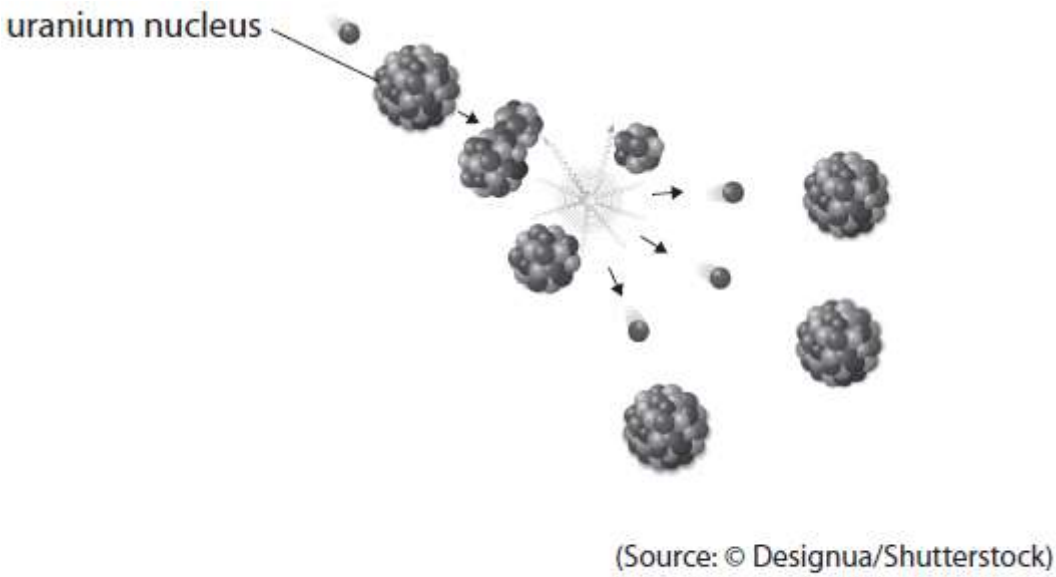

Figure 21

Explain what happens when **one** neutron interacts with **one** uranium nucleus.

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(Total for question = 3 marks)

Q7.

* Radioactive isotopes can be used to investigate cancer and other illnesses.

The thyroid gland in the neck absorbs most of the iodine that our bodies need.

A person can become ill if their thyroid absorbs too little iodine.

Explain how a radioactive isotope with suitable properties may be used to investigate the uptake of iodine by this gland.

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(Total for question = 6 marks)

Q8.

The teacher moves the radiation detector to different distances from the radioactive source.

The teacher determines the mean detector reading at each distance from the source.

The teacher plots the results on graph paper, as shown in Figure 14.

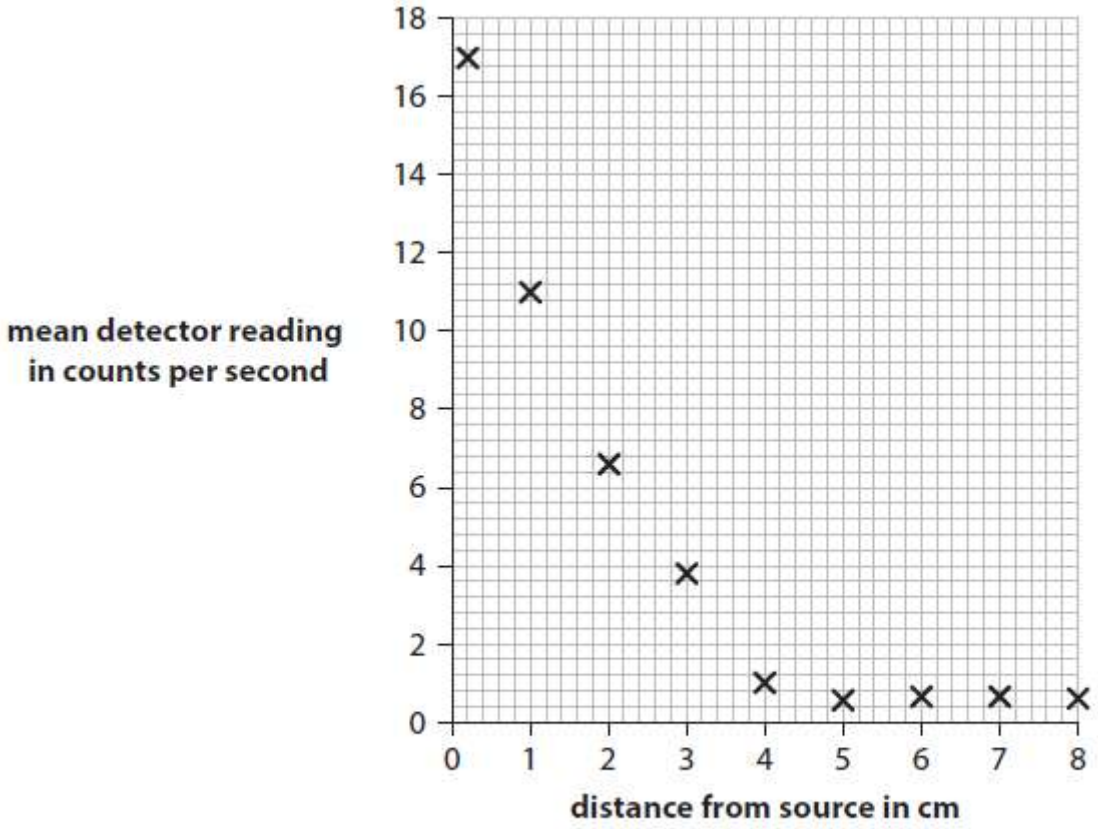

Figure 14

(i) The source emits alpha radiation **only**.

Explain how the graph in Figure 14 shows that the source only emits alpha radiation.

(2)

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(ii) Give a reason why the mean detector reading in Figure 14 does not fall to zero in this experiment.

(1)

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(Total for question = 3 marks)

Q9.

Figure 22 shows a PET scanner used to detect cancerous tumours.

A radioactive isotope is injected into a patient.

The isotope is absorbed by the tumour.

The isotope emits positrons from the location of the tumour.

The ring of radiation detectors rotates around the person.

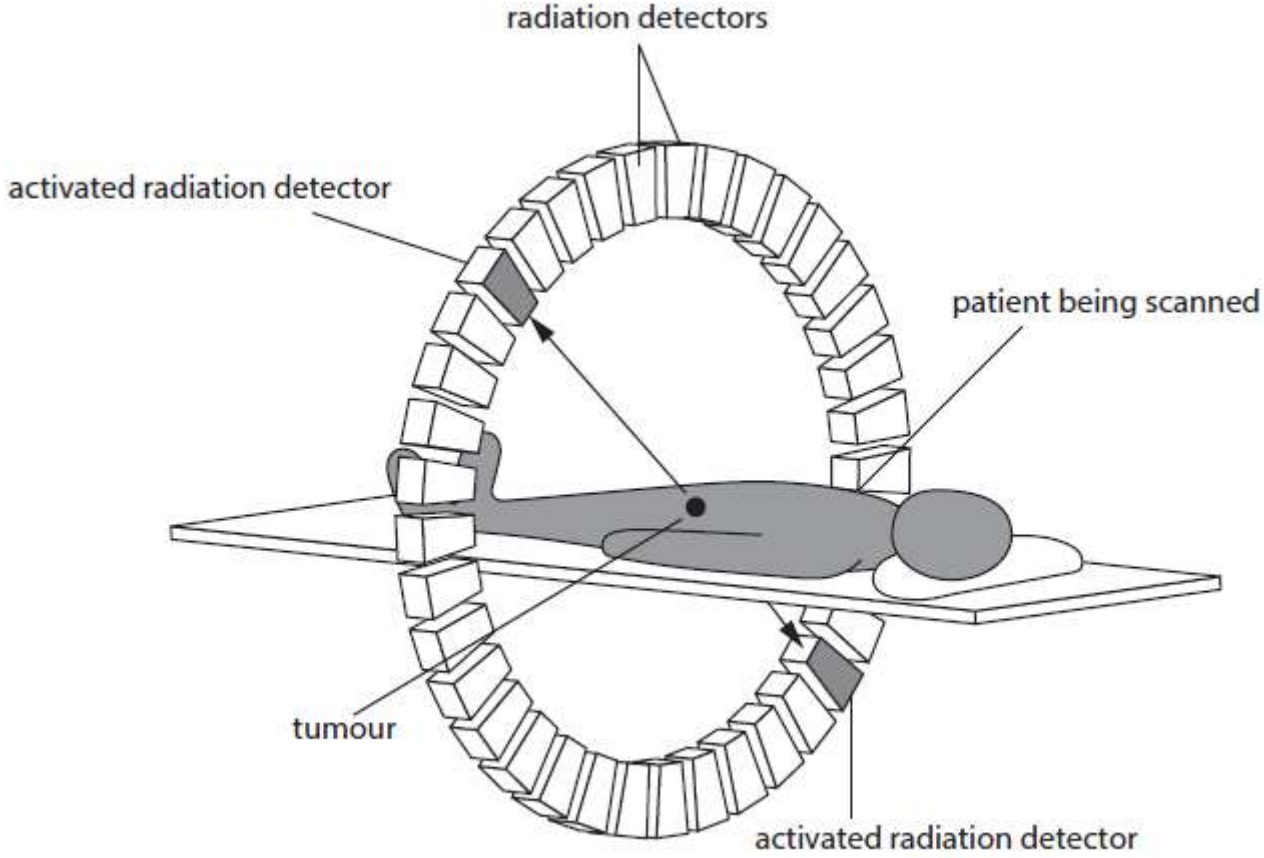

Figure 22

(i) Explain how the scan can give the location of the tumour.

(3)

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(ii) Explain why the radioactive isotope injected into the patient must be produced near to the place where it is to be used.

(2)

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(Total for question = 5 marks)

Q10.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

An atom has a central nucleus containing neutrons and protons.

Electrons orbit the nucleus.

(i) Which row of the table gives the relative mass and charge of a proton?

(1)

		relative mass	charge
☐	**A**	0	+1
☐	**B**	0	−1
☐	**C**	1	+1
☐	**D**	1	−1

(ii) An atom has a radius of 1.0×10^{-10} m.

A nucleus has a radius of 1.0×10^{-15} m.
Calculate the ratio of the radius of the atom to the radius of the nucleus.

(2)

ratio of radius of atoms to radius of nucleus = ...

(iii) Explain why an atom has no charge overall.

(2)

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(Total for question = 5 marks)

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
(i)	An explanation to include; there is no aluminium to absorb β particles (1) (therefore) more β particles reach the G-M tube (1)	 aluminium absorbs/stops/blocks beta particles accept reverse arguments accept radiation for beta particles	(2) AO2

Question number	Answer	Additional guidance	Mark
(ii)	(idea of) background radiation	a named source of background radiation	(1) AO3

Question number	Answer	Additional guidance	Mark
(iii)	becquerel	accept Bq accept close spelling	(1) AO1

Q2.

	Answer	Acceptable answers	Mark
	chain reaction needs a neutron from one fission to reach another uranium nucleus/atom (at the right speed) (1) (fission of 238) needs fast/high(er) energy neutrons (1)	idea of continuous nature of chain reaction the neutrons would be going too slowly /do not have enough energy / lose energy too fast	**(2)**

Q3.

Question number	Answer	Additional guidance	Mark
	an explanation linking any **two** from		**2** **AO3.3**
	(their) laboratory (equipment) is unlikely to be able to produce required conditions (1)	requires much more equipment than available in a typical laboratory	
	need (very) high temperatures (1)	high (particle) speed / KE	
	need (very) high pressure (1)	high (particle) [density	
	results had not been peer reviewed (1)	not checked by other scientists	
	no other scientist produced similar results (1)		

Q4.

Question number	Indicative content	Mark
	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO3 strand 1 (3 marks)** • americium emits alpha • cobalt emits gamma • americium has long(est) half-life • actinium has short(est) half-life **AO2 strand 1 (3 marks)** • alpha is the better ioniser • gamma is weakly ionising • alpha allows electricity to flow • alpha cannot get out of detector • gamma can get out of detector • americium won't run out • actinium would run out	**(6)**

Level	Mark	Descriptor
	0	• No awardable content
Level 1	1–2	• Interpretation and evaluation of the information attempted but will be limited with a focus on mainly just one variable. Demonstrates limited synthesis of understanding. (AO3) • The explanation attempts to link and apply knowledge and understanding of scientific ideas, flawed or simplistic connections made between elements in the context of the question. (AO2)
Level 2	3–4	• Interpretation and evaluation of the information on both variables, synthesising mostly relevant understanding. (AO3) • The explanation is mostly supported through linkage and application of knowledge and understanding of scientific ideas, some logical connections made between elements in the context of the question. (AO2)
Level 3	5–6	• Interpretation and evaluation of the information, demonstrating throughout the skills of synthesising relevant understanding. (AO3) • The explanation is supported throughout by linkage and application of knowledge and understanding of scientific ideas, logical connections made between elements in the context of the question. (AO2)

SUMMARY, for guidance			
Level	**Mark**	**Additional Guidance**	**General additional guidance – the decision within levels** e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance Elements of physics, i.e isolated fact from knowledge or table	Possible candidate responses americium emits alpha OR alpha needed for smoke detector
Level 2	3–4	Additional guidance Some understanding shown, i.e. link between fact from table and fact from knowledge	Possible candidate responses americium emits alpha that can't get out of the detector **OR** two facts (from either table of knowledge) americium emits alpha and has the long(est) half-life
Level 3	5–6	Additional guidance Understanding is detailed and fully developed, i.e. link between fact from table and fact from knowledge **AND** second fact (from either table or knowledge	Possible candidate responses americium emits alpha that can't get out of the detector AND americium has the long(est) half-life

Q5.

Question Number	Answer	Mark
	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO3 and AO2 (6 marks)** **AO3** • most go straight through to P • some are deflected through small angles to Q • few have deflections greater than 90° to R • or are even reflected (bounce back off the foil) to R **AO2** • alpha positive is repelled by positive nucleus • atom being mostly empty space • atoms have a small nucleus • nucleus has a big mass / density • +ve charge concentrated into a very small space	**(6)**

Level	Mark	Descriptor
	0	• No awardable content
Level 1	1–2	• Interpretation and evaluation of the information attempted but will be limited with a focus on mainly just one variable. Demonstrates limited synthesis of understanding. (AO3) • The explanation attempts to link and apply knowledge and understanding of scientific ideas, flawed or simplistic connections made between elements in the context of the question. (AO2)
Level 2	3–4	• Interpretation and evaluation of the information on both variables, synthesising mostly relevant understanding. (AO3) • The explanation is mostly supported through linkage and application of knowledge and understanding of scientific ideas, some logical connections made between elements in the context of the question. (AO2)
Level 3	5–6	• Interpretation and evaluation of the information, demonstrating throughout the skills of synthesising relevant understanding. (AO3) • The explanation is supported throughout by linkage and application of knowledge and understanding of scientific ideas, logical connections made between elements in the context of the question. (AO2)

Summary for guidance			
Level	**Mark**	**Additional Guidance**	**General additional guidance – the decision within levels** Eg - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance unlinked statement from the diagram or table or knowledge of the atom	Possible candidate responses most particles go to P *(from table)* OR particles refract/bend to Q *(from diagram)*
Level 2	3–4	Additional guidance One link between any TWO of diagram, table, knowledge about atoms.	Possible candidate responses Most particles go straight through (the gold) to P *(from table and diagram)* OR Most particles go to P which means an atom is mainly space *(from table and knowledge)* OR particles are reflected because there is a nucleus (*diagram and knowledge)*
Level 3	5–6	Additional guidance One link between diagram **AND** table **AND** knowledge about atoms	Possible candidate responses Most particles go straight through (the gold) to P which means an atom is mainly space OR A few particles reflected back to R which means an atom has a nucleus

Q6.

Q7.

Question number	Indicative content	Mark
*	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO2 (6 marks)** • use a radioactive isotope of iodine as this is taken up by the gland • isotope given by injection or orally • gland is in the neck, so cannot use an alpha emitter as alpha will not exit through the skin • use beta or gamma emitter • isotope has to have a short enough half-life to minimise exposure to radiation but long enough for the reading to be taken • allow time for isotope to reach gland • use Geiger-Müller tube and counter to determine count rate of isotope in gland • compare with normal count rate to determine whether uptake of iodine is normal	**(6)**

Level	Mark	Descriptor
	0	No awardable content.
1	1–2	• The explanation attempts to link and apply knowledge and understanding of scientific ideas, flawed or simplistic connections made between elements in the context of the question. (AO2) • Lines of reasoning are unsupported or unclear. (AO2)
2	3–4	• The explanation is mostly supported through linkage and application of knowledge and understanding of scientific ideas, some logical connections made between elements in the context of the question. (AO2) • Lines of reasoning mostly supported through the application of relevant evidence. (AO2)
3	5–6	• The explanation is supported throughout by linkage and application of knowledge and understanding of scientific ideas, logical connections made between elements in the context of the question. (AO2) • Lines of reasoning are supported by sustained application of relevant evidence. (AO2)

Q8.

Question number	Answer	Additional guidance	Mark
(i)	an explanation linking any **two** from readings fall (to almost zero) (1) radiation is (all) absorbed (1) after a few cm (of air) (1)	 accept graph / activity / measurements for readings stopped by air in a short distance (in air) reverse arguments must include beta **and** gamma	**2** **AO2.1**

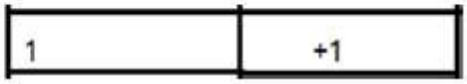
Question number	Answer	Additional guidance	Mark
(ii)	background radiation (1)	or words to that effect accept named examples ignore reference to original alpha source	1 AO2.1

Q9.

Question	Answer	Additional guidance	Mark
(i)	an explanation linking any 3 from positrons and electrons annihilate (1) (two) gamma rays produced/emitted (1) in opposite directions /at 180° (1) detected by radiation detector/ gamma cameras/scintillation counters (1) at (almost) the same time (1) time difference gives distance difference (1)	ignore positrons for this marking point allow triangulation	3 AO2.2

Question	Answer	Additional guidance	Mark
(ii)	an explanation linking any two from must be used a short time after production (1) half-life is short (1)	must be used while activity is high	2 AO1.2
	activity decreases rapidly/decays rapidly (1)	accept decays before use / does not last long	

Q10.

Question Number	Answer	Mark
(i)	C  **A** is incorrect the proton has a mass of 1 not 0 **B** is incorrect the proton has a mass of 1 not 0 **D** is incorrect the proton has a charge of +1 not -1	(1) AO1

Question Number	Answer	Additional guidance	Mark
(ii)	substitution (1) ratio = $\frac{10^{-10}}{10^{-15}}$ evaluation (1) 10^5	$10^{-10} : 10^{-15}$ accept suitable equivalent ratios e.g. $1 \times 10^5 : 1$ $1 : 10^{-5}$ or $10^5 : 1$ 1: 0.00001 or 100000:1 allow 1 mark for inverted ratios e.g. $10^{-15} : 10^{-10}$ 0.00001:1 or 1:100000 award full marks for the correct answer with no working	(2) AO2

Question Number	Answer	Additional guidance	Mark
(iii)	an explanation linking same number / amount of (1) electrons and protons (1)	 equal number / amount of allow balanced (number / amount of) negative and positive charges ignore (neutral) neutrons reject positive/negative neutrons for 2nd marking point	(2) AO1